

High Friction Leg Tip Design for Walking Hexapod Robot: Design Process and Performance Evaluation

Grzegorz Lange

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Abstract

This document presents the design process and performance evaluation of leg tip attachments for a walking hexapod robot. The primary objective was to address traction issues encountered on smooth and slippery surfaces through the development and testing of various rubber tip designs. Five different tip geometries were designed, manufactured, and tested: spherical base, flat base spherical, lattice spherical, flat with flower petals, and a hybrid design combining petals and lattice structures. Testing revealed that material selection is as critical as geometric design, leading to the development of silicone-based tips with TPU cores for optimal performance across various terrains.



1 Introduction

Walking robots operating in diverse environments face significant challenges when encountering surfaces with varying friction characteristics. Hexapod robots, in particular, rely heavily on effective ground contact through their six legs to maintain stability and locomotion. This research addresses the critical issue of traction loss on smooth and slippery surfaces through systematic design and evaluation of specialized leg tip attachments.

1.1 Problem Statement

The hexapod robot experiences significant traction difficulties when operating on smooth surfaces such as polished floors, laminated surfaces, and other low-friction materials. These problems are amplified when dealing with small servomotors which are unable to maintain target position under heavy load. These conditions can lead to the following:

- Loss of locomotion control
- Reduced walking stability
- Inability to navigate certain terrains effectively
- Compromised performance of contact detection systems

1.2 Objectives

The primary objectives of this research are:

1. Design multiple leg tip geometries to improve surface traction
2. Evaluate performance across different surface types and conditions
3. Develop an optimal solution for indoor and outdoor applications

2 Design Methodology



2.1 Design Concepts

Five distinct leg tip designs were developed and evaluated:

1. **Spherical Base:** Traditional rounded tip design serving as baseline
2. **Flat Base Spherical:** Flattened contact surface with spherical edges
3. **Lattice Spherical:** Spherical design with external lattice structure for deformation
4. **Flat Flower Petals:** Flat base with petal-like protrusions for increased contact area and adapt to angular changes
5. **Hybrid (Petals + Lattice):** Combined design incorporating both petal and lattice features

2.2 Material Selection

Two primary materials were investigated:

- **TPU (Thermoplastic Polyurethane):** Initial material choice for 3D printing capability. It is a soft material with great elastic deformation and a high coefficient of friction (COF).
- **Silicone:** Alternative material offering superior grip characteristics and high tackiness as a result of its high surface energy.

2.3 Manufacturing Process

2.3.1 TPU Components

TPU tips were manufactured using 3D printing technology, allowing complex internal geometries such as lattice structures and cavities. Custom Gcode approaches and advanced slicer features allowed for manufacturing exposed/external lattice structures creating textured surface for improved grip.

2.3.2 Silicone Components

Silicone tips were produced using custom molds for the external silicone lining combined with TPU cores for structural integrity. The manufacturing process involved

1. Mold design and fabrication
2. TPU core insertion for attachment interface
3. Silicone casting and curing

3 Testing and Evaluation

3.1 Testing Methodology

Performance evaluation was performed through single-leg testing on multiple types of surface. Each design was assessed using the following criteria:

- **Deformation:** Ability to conform to surface irregularities
- **Contact Area:** Maintenance of contact area under angular displacement
- **Vertical Contact Area:** Surface area in direct vertical contact
- **Obstacle Hooking:** Risks of engaging with surface features
- **Terrain Penetration:** Capability to penetrate soft surfaces and potential risks of jamming the legs into the ground.
- **Shock Absorption:** Energy dissipation during impact
- **Dust/Sand Trap:** Tendency to accumulate debris

3.2 Surface Types

Testing was carried out on the following surfaces:

- Smooth laminate
- Paper
- Hard foam
- Porous laminate

Due to the lack of professional testing equipment, friction performance was rated on a scale: low < medium < decent < good < excellent.

3.3 Shape Testing Results

Table 1: Design Characteristics and Mechanical Properties

Design	Deformation	Contact Area/Angle	Vertical Contact Area	Obstacle Hooking	Terrain Penetration	Shock Absorption	Dust/Sand Trap
Spherical Base	Decent	Maintains same	Small	Low	Medium	Medium	No
Flat Base Spherical	Low	Low past certain angle	Large	Low	Low	Low	No
Lattice Spherical	Big	Maintains similar	Medium	Low	Medium	High	Yes
Flat + Flower Petals	Medium	Slightly decrease	Large	Medium/Low	Low	Low	No
Hybrid	Big	Excellent (remains flat when tilted)	Large	Low	Low	Medium/Low	Yes

Table 2: Surface Performance Comparison Across Different Materials

Design	Laminate Smooth	Paper	Hard Foam	Laminate Porous
Spherical Base	Low	Medium	Medium	Decent
Flat Base Spherical	Decent	Decent	Medium	Decent
Lattice Spherical	Low/Medium	Medium	Decent	Decent
Flat + Flower Petals	Decent	Decent	Medium	Decent
Hybrid	Decent	Decent	Medium	Medium

3.4 Shape Testing Key Findings

3.4.1 Flat Base

Flat base approach greatly increases the contact surface area, however it struggles with maintaining it even at shallow angles.

3.4.2 Flat Base with Flower Petals

Flower petals were added to directly address the issue with maintaining the contact surface area. The idea behind the flower petals is that, once the angle increases the flower petal will drag and deform the main tip surface.

3.4.3 Lattice structures

Incorporating the external lattice structure resulted in improvements in both traction and deformation. The open lattices however have a tendency to accumulate sand and derbies, which results in sharp decline in performance. This approach is only suitable for clean environments and possible only using 3D printing technology.

3.4.4 Hybrid Design

The hybrid design combining petals and lattice structure demonstrated superior performance in maintaining contact area under angular displacement while providing adequate deformation characteristics.



3.4.5 TPU Material Limitations

Despite geometric optimizations, TPU material exhibited insufficient grip on smooth surfaces, particularly on laminated floors where the robot continued to experience significant slipping.

3.4.6 Vertical Shock Absorption

The tip designs were engineered to exhibit poor vertical shock absorption performance. This approach was chosen to ensure predictability, since our Hexapod relies on shock absorbing springs coupled with contact detection switches. Allowing too much vertical compression of the tip could result in it getting twisted or sprained and losing balance.

4 Material Transition and Optimization

4.1 Silicone Implementation

Based on testing results indicating material limitations, the design process transitioned to silicone-based tips. Key advantages of silicone include:

- Superior surface grip compared to TPU
- Enhanced deformation characteristics
- Better conformability to surface irregularities
- Reduced slipping on smooth surfaces

4.2 Hybrid Material Approach

The final design incorporates a hybrid material strategy:

- **Silicone exterior:** Provides optimal grip and deformation
- **TPU core:** Ensures structural stability and provides reliable attachment interface
- **Elimination of vertical compression:** TPU core prevents excessive deformation under load

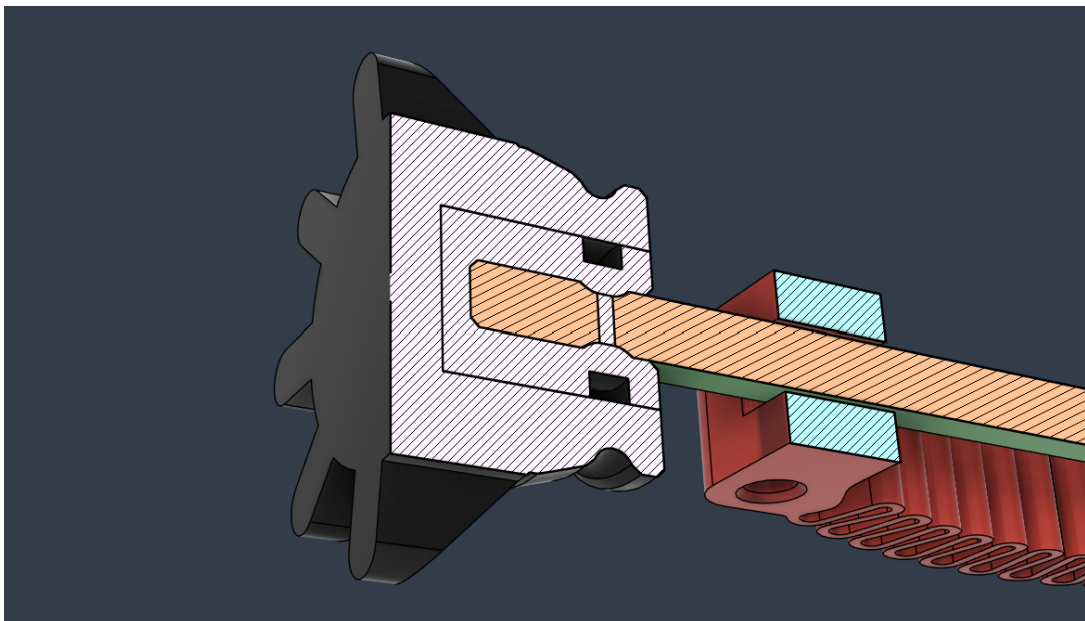
4.3 Design Evolution

The silicone implementation underwent iterative improvement:

1. Initial silicone molding based on hybrid TPU design
2. Volume optimization: Second generation increased to 215% of original volume, for reliability the contact area should be as big as possible
3. Edge rounding to prevent stress concentration and cracking of the cast
4. Improved mold design for consistent manufacturing

4.4 Final Design

The final design consists of flat bottom surface with radially distributed petals structures that are connected via solid lofts to the main shaft. The tip has internal TPU core that extends up to 80 percent of the body. The design maximizes the contact area and conforms to the surface.



4.5 Final Design Tests

The design showed excellent friction performance on all of the tested surfaces. The TPU core decreased the vertical deformation, and it was rated as "Medium" compared to other designs. The contact area was large and the main body of the tip is capable of maintaining its vertical position with relation to the ground for shallow angles. For larger angles the contact area decreases only slightly. The risks of terrain penetration and obstacle hooking are minimal, Dust and sand might stick to the surface, but is easy to clean.

5 Results and Discussion

5.1 Performance Validation

Testing confirmed the theoretical predictions regarding material performance:

- TPU designs, despite geometric optimization, remained insufficient for smooth surface traction
- Silicone implementation dramatically improved grip performance
- Hybrid material approach successfully balanced grip and structural integrity

5.2 Design Trade-offs

The evaluation revealed several important trade-offs:

- **Deformation vs. Stability:** Higher deformation improves surface conformity but may compromise structural stability
- **Contact Area vs. Angular Performance:** Larger flat surfaces provide better vertical contact but may lose effectiveness at angles
- **Grip vs. Debris Accumulation:** Designs with textures provide better grip characteristics but tend to trap more dust and sand

6 Conclusions and Possible Improvements

6.1 Conclusions

1. The hybrid design combining petals and lattice structure provides optimal geometric characteristics for indoor applications
2. Material selection is equally critical as geometric design for overall performance
3. Silicone material significantly outperforms TPU in terms of surface grip
4. Hybrid material approach (silicone exterior with TPU core) offers the best balance of performance characteristics
5. Develop surface-specific tip variants for specialized applications

6.2 Future Work

Future research directions include:

- Designs with surface dimples and or protrusions
- Designs with variable suction cups
- Investigation of smart materials for adaptive grip characteristics
- Integration of sensory capabilities within tip structures
- Long-term durability testing under operational conditions